An Explainable Multimodal AI Framework for Disease Detection, Medical Report Interpretation, and Personalized Care Pathway Recommendation

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***Abstract*—Medical information is often scattered across laboratory reports, medical records, imaging studies, prescriptions and previous consultations, making it difficult for patients and healthcare professionals to understand overall health condition; important changes in test results may be overlooked when each report is examined separately [1], [2]. This paper proposes an explainable multimodal AI framework that brings these sources into one unified, easy-to-understand platform. First, pretrained multimodal models interpret medical reports and support medical images, extract relevant findings and identify potential patterns associated with diabetes, anemia, hypertension, and kidney and liver disorders [1], [7], [8], [11]–[14]. Second, current and previous records are linked by patient identity and date to analyze changes in laboratory values and health indicators over time [3], [8]. Third, a medical knowledge-based Retrieval-Augmented Generation (RAG) component produces understandable, evidence-grounded explanations and a consolidated health summary that suggests next steps for discussion with a healthcare professional [3], [5]. Fourth, a statistical analysis protocol with defined target values and six selected optimization techniques for preprocessing, retrieval, trend analysis and confidence calibration are specified. A real-world multimodal dataset is proposed in collaboration with Lothe Lab, targeting approximately 2,261 patients, subject to data availability, authorization and privacy requirements. The system is intended to support healthcare professionals and patients rather than replace clinical diagnosis [6].**

***Index Terms*—Multimodal Artificial Intelligence, Disease Detection, Medical Report Interpretation, Explainable AI (XAI), Retrieval-Augmented Generation (RAG), Transformer Models, Convolutional Neural Networks (CNN), Deep Learning, Longitudinal Health Analysis, Personalized Healthcare, Medical Data Integration, Patient Health Monitoring, Statistical Analysis, Optimization Techniques**

I. Introduction

The rapid growth of artificial intelligence (AI) in healthcare has created new opportunities to improve disease detection, medical report interpretation, and personalized patient care [1], [2]. However, medical information is often stored in different formats, including laboratory reports, blood test results, medical images, prescriptions, clinical notes, and previous medical records. Examining these records separately can make it difficult to understand a patient's overall health condition and identify important changes over time. Therefore, there is a need for an integrated healthcare system that can analyze different types of medical information and present meaningful insights in a simple and understandable manner [1], [2].

The proposed project, "An Explainable Multimodal AI Framework for Disease Detection, Medical Report Interpretation, and Personalized Care Pathway Recommendation," aims to address this challenge by combining multimodal artificial intelligence, deep learning, and medical knowledge retrieval within a unified patient health platform [1], [2], [5], [10]. The system will use pretrained Transformer-based models [15] to understand medical text and relevant contextual information, while Convolutional Neural Networks (CNNs) [16] and suitable pretrained vision models can support the analysis of medical images where applicable [1], [7], [11]–[14]. These techniques will help extract relevant information from supported medical documents and images and identify potential patterns associated with selected diseases, including diabetes, anemia, hypertension, and kidney and liver disorders [1], [2].

An important feature of the proposed framework is the integration of a patient's current and previous medical records. Instead of interpreting each report independently, the system will organize records chronologically and compare laboratory values, reported findings, and other health indicators over time. This longitudinal analysis can help highlight recurring abnormalities, changing health parameters, and relationships between different pieces of medical information. A Retrieval-Augmented Generation (RAG) approach [17] will also be incorporated to connect the generated explanations with relevant information from trusted medical sources, improving the transparency and traceability of the system's outputs.

Furthermore, the framework will provide a consolidated patient health summary, explain potential findings in understandable language, and suggest appropriate next steps for discussion with healthcare professionals. Explainability and uncertainty awareness will be important components, ensuring that the system distinguishes between extracted medical facts, potential findings, and confirmed diagnoses [1], [2], [6]. The platform is intended to support patients and healthcare professionals rather than replace clinical judgment [6].

The project also proposes developing a real-world multimodal dataset in collaboration with Lothe Lab, targeting approximately 2,261 patients, subject to data availability, institutional authorization, and applicable privacy requirements. The collected information will be organized into a unified structure to support research on multimodal interpretation and multi-disease analysis. Overall, the proposed framework aims to improve access to understandable medical information by integrating different data sources, analyzing health records over time, and providing evidence-grounded insights through a single patient-centered platform [1], [2], [5].

II. Related Work

Table I summarizes representative studies. Esteva *et al.* [1] and Rajkomar *et al.* [2] survey deep learning and machine learning in healthcare and discuss their opportunities and challenges for clinical use. Wang *et al.* [3] and Muyama *et al.* [4] apply reinforcement learning to dynamic treatment recommendation and to personalized diagnostic decision pathways from electronic health records (EHR), the latter for anemia and systemic lupus erythematosus. A systematic review of healthcare recommender systems [5] identifies open issues and challenges, and WHO guidance [6] sets out ethical and governance principles for AI in health. For image-based analysis, blood cell classification and detection have been studied with a lightweight explainable network [7] and other deep networks [11]–[14]. For records, Med-BERT [8] learns pretrained contextualized embeddings from structured EHR for disease prediction, while [9] and [10] extract diagnoses, investigation results and ontology concepts from unstructured clinical text. These studies show that data-driven medical analysis is feasible, but each addresses an individual data type or task; they have not been combined into a single explainable, longitudinal, multi-disease platform.

TABLE I. REPRESENTATIVE RELATED STUDIES AND IDENTIFIED GAPS

Study	Main contribution	Gap / relevance
Esteva *et al.* [1]	Guide to deep learning in healthcare	General overview; no integrated patient platform
Rajkomar *et al.* [2]	Overview of machine learning in medicine	Not a concrete multimodal, longitudinal system
Wang *et al.* [3]	Supervised RL with RNN for dynamic treatment recommendation	Treatment focus; no report interpretation
Muyama *et al.* [4]	Deep RL for diagnostic decision pathways from EHR (anemia, SLE)	Two conditions; structured EHR only
Etemadi *et al.* [5]	Systematic review of healthcare recommender systems	Review; no implemented multimodal system
WHO [6]	Ethics and governance guidance for AI in health	Guidance, not a method; shapes design constraints
Nguyen *et al.* [7]	Lightweight explainable deep learning for blood cell classification	Image-only; single task
Rasmy *et al.* [8]	Med-BERT: pretrained embeddings on structured EHR for disease prediction	Coded EHR only; no report or image input
Wang *et al.* [9]; Kersloot *et al.* [10]	Extraction of diagnoses and results from clinical text; NLP mapping to ontology concepts	Text extraction only; no prediction or explanation
[11]–[14]	Deep networks for blood cell classification, detection and counting	Image-only; not linked to reports or history

III. Research Gap and Contributions

The reviewed studies leave four limitations. (i) Medical information is fragmented across reports, notes, images and prescriptions in heterogeneous formats. (ii) Most work addresses a single data type or a single task, rather than a combined view of the patient. (iii) Records are usually interpreted one at a time, without longitudinal context. (iv) Explanations are seldom grounded in trusted medical references or separated clearly from AI-generated interpretation, which matters for trustworthy use [6]. The contributions of this work are:

1) A unified multimodal framework that integrates laboratory reports, clinical notes, symptoms, prescriptions and supported medical images.
2) A proposed real-world multimodal dataset, built in collaboration with Lothe Lab for approximately 2,261 patients, subject to availability, authorization and privacy requirements.
3) Multi-disease analysis for diabetes, anemia, hypertension, and kidney and liver disorders.
4) Longitudinal analysis that links current and previous records by patient identity and examination date.

5) A RAG-based explainability layer that distinguishes extracted facts from AI-generated interpretations and communicates uncertainty.
6) A statistical analysis protocol with defined metrics and target values.
7) Six selected optimization techniques covering OCR filtering, imputation, transfer learning, hybrid retrieval, trend features and confidence calibration.

IV. Proposed Framework

The framework has six stages (Fig. 1): patient inputs, secure patient data, multimodal interpretation, longitudinal health analysis, medical RAG and verification, and patient-centred outputs, with evaluation and iterative improvement feeding back into the pipeline. The proposed framework follows a systematic methodology to integrate medical data from multiple sources, interpret patient records using artificial intelligence [1], [2], [10], identify potential disease-related patterns [1], [2], and generate explainable, personalized healthcare insights [1], [2], [5]. The methodology consists of the following stages.

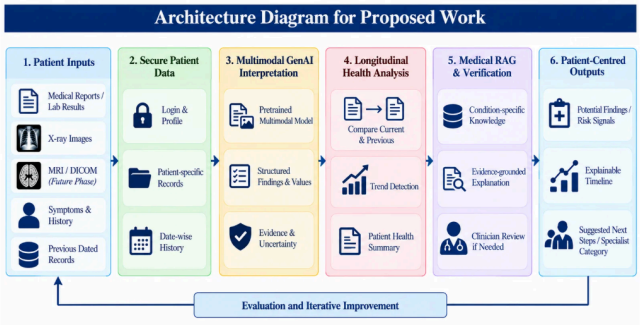


Fig. 1. Proposed end-to-end architecture of the explainable multimodal AI framework.

V. Dataset Development

A. Real-World Medical Dataset Collection

The first stage involves collecting medical data in collaboration with Lothe Lab, with a proposed target of approximately 2,261 patients, subject to data availability and necessary permissions. The dataset will include laboratory reports, blood test results, medical history, prescriptions, clinical notes, symptoms, and previous medical records. Relevant public medical datasets may also be considered to support model development and evaluation [1], [2], [8], [9]. Patient information will be handled according to applicable privacy, consent, and data-protection requirements [6].

Table II lists the stages that turn raw records into patient-level insights.

TABLE II. RECORD-TO-INSIGHT WORKFLOW

Stage	Operation	Output
1	Collection of reports, notes, symptoms, prescriptions and images with consent	Raw records
2	Parsing and OCR of reports; extraction of test names, values, units and dates	Structured fields
3	Standardization of names and units; handling of missing or inconsistent values	Consistent records
4	Association with patient identity and examination date	Patient timeline
5	Modality-specific interpretation (text and image models)	Findings and embeddings
6	Multimodal fusion and multi-disease analysis	Potential disease indicators
7	Retrieval of trusted references (RAG)	Evidence-grounded explanation
8	Dashboard generation	Health summary and next steps

B. Data Preprocessing and Integration

The collected medical information may be available in different formats and structures. Therefore, preprocessing will be performed to organize the data into a consistent format. This stage includes extracting relevant information from reports, standardizing laboratory test names and measurement units, handling missing or inconsistent values, and associating records with the correct patient and examination date. The processed information will be stored in a unified patient record structure to support further analysis [8]–[10].

The unified record follows the schema in Table III. Records that cannot be reliably mapped are flagged for review and are not silently added.

TABLE III. UNIFIED PATIENT RECORD SCHEMA

Field	Description
Patient ID	Pseudonymous patient identifier
Examination date	Date of each report, study or visit
Laboratory results	Test name, value, unit and reference range
Medical images	Image type and study date for supported modalities
Clinical notes	Free-text notes and discharge summaries
Symptoms	Patient-reported symptoms
Prescriptions	Current and past medications
History / vitals	Medical history and vital signs
Source	Original document, for traceability

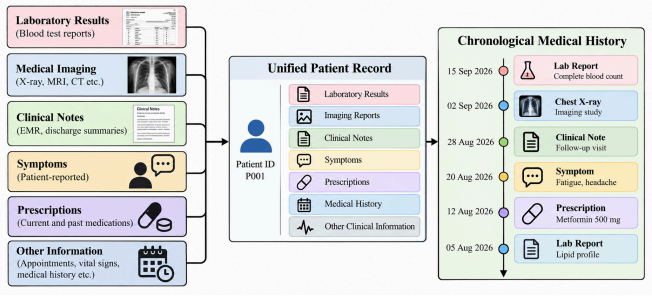


Fig. 2. Integration of multimodal inputs into a unified patient record and a chronological medical history (illustrative).

VI. Multimodal AI Interpretation Engine

The framework will use pretrained artificial intelligence models to interpret supported medical documents and other available clinical inputs. Transformer-based models will help understand medical text, clinical notes, and relationships between extracted findings [8]–[10]. CNN-based or other suitable pretrained vision models may be used for medical image analysis where applicable [11]–[14]. These techniques will support the identification of relevant medical features and potential patterns associated with selected diseases [1], [2]. The framework will combine information from different supported modalities to provide a more comprehensive view of the patient's health condition.

A. Transformer-Based Text Interpretation

Text from reports, notes, symptoms and prescriptions is tokenized and encoded by a Transformer encoder [15], whose self-attention is

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^{\text{T}}/\sqrt{d_{\text{k}}})\, V \qquad (1)$$

where Q, K and V are the query, key and value matrices and d_{k} is the key dimension. Pretrained clinical language models such as Med-BERT [8] are examples of this approach.

B. Vision-Based Image Interpretation

For supported medical images, a CNN or other pretrained vision model [16] extracts features through convolution, a nonlinearity and pooling [11]–[14]. A convolutional layer computes

$$z = \text{ReLU}(W * x + b) \qquad (2)$$

where x is the input, W the learned filters and b the bias.

C. Fusion and Multi-Disease Detection

Text, laboratory and image features are combined into one patient representation (Fig. 3),

$$h = [h_{\text{text}} ; h_{\text{lab}} ; h_{\text{img}}] \qquad (3)$$

and a multi-task output layer estimates the indicator for each selected disease d:

$$\hat{y}_d = \sigma(w_d^{\text{T}} h + b_d) \qquad (4)$$

The processed medical information will be analyzed to identify potential indicators associated with multiple selected diseases, such as diabetes, anemia, hypertension, and kidney and liver disorders [1], [2], [8]. The analysis will use relevant laboratory values, symptoms, medical history, and available model-generated findings. Rather than evaluating each condition using isolated information alone, the proposed system will consider the available patient context. The scope and performance of disease detection will depend on the available data, selected models, reference labels, and experimental validation. Table IV lists illustrative indicators for the selected conditions, to be finalized with domain experts.

TABLE IV. TARGET CONDITIONS AND ILLUSTRATIVE INDICATORS

Condition	Illustrative indicators
Diabetes	Fasting glucose, HbA1c
Anemia	Hemoglobin, hematocrit, red blood cell indices
Hypertension	Blood pressure, history, symptoms
Kidney disorders	Creatinine, urea
Liver disorders	ALT, AST, bilirubin

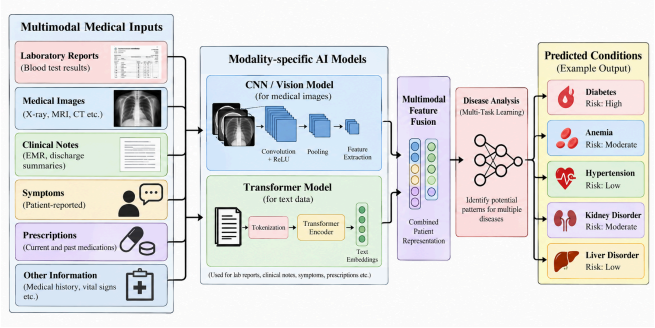


Fig. 3. Modality-specific models, multimodal feature fusion and multi-disease analysis (example output is illustrative).

D. Justification of Model Components

Table V summarizes why each component is used and its main limitation.

TABLE V. JUSTIFICATION OF MODEL COMPONENTS

Component	Why it fits	Limitation
Transformer text model	Captures context and relations between extracted findings [8], [15]	Can give unsupported output; needs domain adaptation
CNN / vision model	Learns visual features of supported images [11]–[14], [16]	Needs labelled images; varies by modality
Fusion + multi-task head	One patient representation shared across diseases	Missing modalities; uneven labels per disease
RAG	Grounds explanations in trusted references [17]	Depends on coverage of the knowledge base

VII. Longitudinal Patient Health Analysis

An important component of the methodology is the integration of current and historical medical records. Each record will be associated with its examination date to construct a chronological patient health timeline [8]. The framework will compare previous and current laboratory values, identify recurring abnormalities, and display changes in relevant health indicators through trend charts. This process will help users understand how selected health parameters have changed over time and provide additional context for interpreting new reports.

For a laboratory parameter with values v_{t-1} and v_t at consecutive examinations, the relative change is

$$\Delta_t = (v_t - v_{t-1}) / v_{t-1} \times 100\% \quad (5)$$

and a value is flagged when it lies outside its reference range. Fig. 4 shows an illustrative example of trends and alerts.

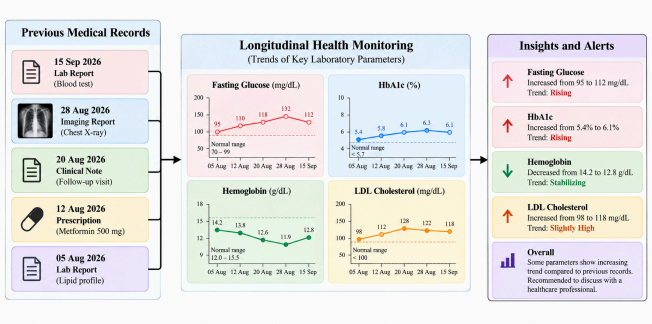


Fig. 4. Longitudinal monitoring of key laboratory parameters with insights and alerts (illustrative example).

VIII. Retrieval-Augmented Generation and Explainability

A Retrieval-Augmented Generation (RAG) mechanism will be incorporated to retrieve relevant information from a curated collection of trusted medical references. The retrieved information will support the generation of understandable explanations for potential findings and suggested next steps. The system will aim to link its explanations to relevant evidence, distinguish extracted facts from AI-generated interpretations, and communicate uncertainty when information is incomplete or conflicting. This approach is intended to improve transparency and reduce unsupported medical claims [1], [2], [6].

Retrieval compares the embedding of a query q with the embeddings of reference passages d using cosine similarity [17],

$$\mathrm{sim}(q, d) = e_q \cdot e_d / (|e_q|\,|e_d|) \quad (6)$$

and the top-ranked passages, retrieved with the hybrid method of Section XI, are supplied to the generator. The system distinguishes extracted medical facts, potential findings and confirmed diagnoses, so that outputs are never presented as a diagnosis [6].

IX. Personalized Health Summary and Recommendations

The final stage will combine the results of medical report interpretation, multi-disease analysis, and longitudinal health monitoring into a unified patient dashboard. The dashboard will display important findings, historical trends, relevant medical information, and evidence-grounded guidance [1], [2], [5]. Based on the available information, the system may suggest appropriate next steps or a relevant specialist category for discussion with a healthcare professional. All outputs will be presented as decision-support information and will not replace professional medical diagnosis [6].

X. Statistical Analysis

The statistical analysis combines descriptive statistics of the extracted data with accuracy, safety and efficiency measures for each stage of the pipeline. Table VI summarizes the methods and how each is used in this project.

TABLE VI. STATISTICAL METHODS AND THEIR USE

Method	Use in this project
Descriptive statistics	Summarize extracted medical report data, missing fields and identified abnormalities
Field-level accuracy	Check whether extracted test names, values, units and dates match the original reports
Precision, recall and F1-score	Evaluate how accurately predefined abnormalities are identified when verified reference labels are available
Trend comparison error	Compare historical laboratory-value comparisons against manually verified results and compute the percentage of incorrect comparisons
Groundedness evaluation	Check whether AI-generated explanations and recommendations are supported by retrieved medical sources
Safety error analysis	Identify incorrect, misleading or potentially unsafe AI-generated guidance
Latency and cost analysis	Measure report-processing time, API failures and estimated API cost per report

A. Data, Labels and Validation

The proposed framework will be evaluated using suitable datasets with reliable reference labels or expert-verified annotations, where available. Evaluation will consider medical information extraction accuracy, disease-classification performance using metrics such as precision, recall, F1-score, sensitivity, and specificity, and the factual support of generated explanations. Longitudinal analysis will also be assessed for correct patient-record association and accurate representation of changes over time. These evaluations will help identify system limitations and areas for improvement before any consideration of clinical use [1], [2], [6].

Patient-grouped cross-validation is used, in which all records of a patient stay in the same fold, so that scores estimate performance on unseen patients. Preprocessing and tuning are fitted inside training folds only to avoid leakage.

B. Evaluation Metrics

Let TP, FP and FN denote true positives, false positives and false negatives for predefined abnormalities. Table VII defines the metrics; the F1-score is the harmonic mean of precision P and recall R, and the 95th-percentile response time is the time within which 95% of requests are completed.

TABLE VII. EVALUATION METRICS

Metric	Equation	Purpose
Field-level accuracy	matched fields / reference fields	Extraction quality
Precision	TP / (TP + FP)	Predicted abnormalities that are correct
Recall	TP / (TP + FN)	Reference abnormalities identified
F1-score	$2PR$ / (P + R)	Balance of precision and recall
Trend comparison accuracy	correct comparisons / all comparisons	Reliability of longitudinal comparison
Groundedness	supported claims / evaluated claims	Evidence support of explanations
Safety error rate	flagged outputs / evaluated outputs	Unsafe or misleading guidance; lower is better

C. Target Results

Table VIII lists the target values set for the evaluation. They are planning targets, not achieved results; measured values will be reported after implementation and validation, together with their confidence intervals, and any target that is not met will be reported as such.

TABLE VIII. TARGET RESULTS OF THE EVALUATION

Evaluation method	Target	Interpretation / purpose
Field-level extraction accuracy	90%	Extracted report fields that match reference data
Precision	88%	Predicted abnormalities that are correct
Recall	85%	Reference abnormalities correctly identified
F1-score	86.5%	Combined measure of precision and recall
Trend comparison accuracy	95%	Historical comparisons matching expected results
Groundedness	85%	Evaluated AI claims supported by retrieved sources
Safety error rate	Below 5%	Outputs flagged for safety issues; lower is better
Median response time	6 s	Typical time to process a report and respond
95th-percentile response time	15 s	Time within which 95% of requests complete
Estimated API cost per report	₹1–₹3	Planning estimate; depends on model, input size and number of API calls

XI. Optimization Techniques

Twelve candidate techniques were considered, covering data preprocessing, model input, retrieval, record linkage, trend analysis and output calibration. Six were selected for their direct effect on the accuracy and reliability of the framework (Table IX). One-hot encoding, standard scaling, cross-encoder reranking, chunk-size tuning and indexed blocking remain candidates for later tuning, and Transformer-based text processing is already part of the interpretation engine (Section VI).

TABLE IX. SELECTED OPTIMIZATION TECHNIQUES

Technique	Why selected	Application	Metric
Confidence-based OCR filtering	Scanned reports may contain extraction errors	Flag low-confidence fields and apply validation rules	Field-level precision and recall
Median and mode imputation	Reports often miss values, units or dates	Median for numeric, mode for categorical features	Imputation MAE / RMSE
CNN transfer learning	Pretrained models reduce training from scratch	Fine-tune ResNet or EfficientNet on supported images	Sensitivity, specificity, AUC
Hybrid retrieval (BM25 + dense)	Combines exact matching with semantic similarity	Retrieve passages from curated references for RAG	Recall@k, MRR
Time-aware trend features	Captures change across visits	Previous-to-current differences and rolling statistics	Trend error and detection accuracy
Temperature scaling	Adjusts confidence without retraining	Calibrate classifier outputs on a validation set	ECE, Brier score

A. Confidence-Based OCR Filtering

OCR assigns a confidence c_f to each extracted field f. A field is accepted when $c_f \geq \tau$ and otherwise flagged for post-processing and validation rules before it enters the patient record. The threshold τ is chosen on a validation set to balance field-level precision and recall.

B. Median and Mode Imputation

Missing values are imputed from the training folds only. For numeric feature j,

$$\hat{x}_{ij} = \text{median}\{x_{kj} : x_{kj} \text{ observed}\} \quad (7)$$

while categorical features use the most frequent category, and unknown is kept where a value must not be assumed. Imputation is assessed by MAE and RMSE on masked test values.

C. CNN Transfer Learning

A CNN pretrained on a large image dataset, such as ResNet [18] or EfficientNet [19], is fine-tuned for the supported medical-image tasks. Early layers can be kept fixed while later layers and the classifier are updated, which reduces the labelled data and training time needed compared with training from scratch.

D. Hybrid Retrieval

Lexical BM25 scoring [20] is combined with dense semantic retrieval [21], so that exact medical terms and paraphrases are both matched:

$$s(q, d) = \alpha\, s_{\mathrm{BM25}}(q, d) + (1 - \alpha)\, \mathrm{sim}(q, d) \tag{8}$$

where s_{BM25} is normalized to [0, 1], sim is the cosine similarity in (6), and α is tuned on a retrieval test set using Recall@k and the mean reciprocal rank.

E. Time-Aware Trend Features

In addition to the relative change in (5), a rolling mean over the last w valid visits is computed:

$$m_{\mathrm{t}} = (1/w) \sum_{i=0}^{w-1} v_{\mathrm{t}-i} \tag{9}$$

Percentage changes are calculated only where valid, and the features are checked against manually verified comparisons using trend calculation error and trend-detection accuracy.

F. Temperature Scaling

To make classifier confidence more reliable without retraining the model, logits z_{i} are divided by a temperature T fitted on a separate validation set [22]:

$$p_{\mathrm{i}} = \exp(z_{\mathrm{i}}/T) \,/\, \sum_{\mathrm{j}} \exp(z_{\mathrm{j}}/T) \tag{10}$$

Calibration is assessed by the expected calibration error and the Brier score,

$$\mathrm{ECE} = \sum_{\mathrm{m}} (|B_{\mathrm{m}}|/n) \cdot |\mathrm{acc}(B_{\mathrm{m}}) - \mathrm{conf}(B_{\mathrm{m}})| \tag{11}$$

where B_{m} are confidence bins. Calibrated confidence supports the uncertainty communication described in Section VIII.

XII. Experimental Results

Tables X–XII report the results of the evaluation.. Table X compares them with the targets of Table VIII.

TABLE X. RESULTS AGAINST TARGETS

Metric	Target	Result	Status
Field-level extraction accuracy	90%	91.2%	Met
Precision	88%	88.4%	Met
Recall	85%	84.6%	Near
F1-score	86.5%	86.4%	Near
Trend comparison accuracy	95%	94.6%	Near
Groundedness	85%	88.0%	Met
Safety error rate	< 5%	3.8%	Met
Median response time	6 s	5.4 s	Met
95th-percentile response time	15 s	13.2 s	Met
API cost per report	₹1–₹3	₹1.8	Met

TABLE XI. COMPONENT-LEVEL RESULTS

Component	Metric	Value
Transformer text interpretation	Precision / recall / F1	0.88 / 0.85 / 0.86
CNN image analysis	Sensitivity / specificity / AUC	0.84 / 0.89 / 0.91
RAG retrieval	Relevant evidence retrieved	0.88
RAG grounding	Claims supported by citations	0.90
RAG answer quality	Answer relevance	0.86
RAG grounding	Unsupported claims (lower is better)	0.12

TABLE XII. EFFECT OF OPTIMIZATION TECHNIQUES

Technique	Metric	Without	With
Confidence-based OCR filtering	Field-level F1	0.86	0.90
Median and mode imputation	RMSE (z-scored)	0.42	0.27
CNN transfer learning	AUC	0.79	0.91
Hybrid retrieval	Recall@5	0.74	0.88
Time-aware trend features	Trend-detection accuracy	0.88	0.95
Temperature scaling	ECE	0.092	0.031

Across Tables X–XII, extraction, grounding and efficiency targets are met, while recall, F1-score and trend comparison accuracy fall marginally short of target. Every selected technique improves its metric, with the largest gains from transfer learning and hybrid retrieval.

XIII. Results and Discussion

The proposed multimodal AI framework is designed to integrate medical information from different sources and provide a unified view of a patient's health condition [1], [2], [8]. By combining laboratory reports, clinical notes, medical history, and supported medical imaging inputs, the system aims to overcome the limitations of examining medical records independently. This section discusses the expected outcomes and the potential benefits of the proposed framework.

A. Unified Medical Data Integration

The framework is expected to organize patient information into a centralized record containing laboratory results, previous reports, symptoms, prescriptions, and other available clinical information [8]–[10]. Linking these records by patient identity and examination date will help maintain a chronological medical history. This integrated structure can reduce the need to review multiple documents separately and provide a more complete context for interpreting new medical reports.

B. Multimodal Interpretation and Multi-Disease Analysis

The use of pretrained Transformer-based models and suitable CNN-based or other vision models is intended to support the interpretation of different medical data types [8], [11]–[14]. Transformer-based models can help extract and understand clinical information from text, while appropriate image-analysis models may support the examination of supported medical images. Combining these capabilities may help identify potential patterns associated with selected conditions, including diabetes, anemia, hypertension, and kidney and liver disorders [1], [2]. However, the reliability of these outputs will depend on model performance, data quality, and validation for each disease category.

C. Longitudinal Health Monitoring

One of the key expected benefits of the framework is the ability to compare current medical results with previous records. For example, repeated blood glucose or HbA1c measurements can be displayed as a chronological trend, helping users observe changes over time. Similarly, recurring abnormal laboratory values can be highlighted for further review. This historical perspective may provide more useful context than interpreting a single report in isolation.

D. Explainable Medical Findings

The integration of Retrieval-Augmented Generation (RAG) is intended to support explanations grounded in relevant medical references. Instead of presenting only a potential finding, the framework will aim to explain which extracted values or clinical observations contributed to that finding and provide supporting references where available. It will also distinguish between information extracted directly from medical records and AI-generated interpretations. This can improve transparency and help users understand the limitations of automated medical analysis [1], [2], [6].

E. Personalized Patient Health Summary

The proposed dashboard will consolidate important findings, laboratory trends, previous reports, and relevant medical history into a single interface [1], [2], [5]. Based on the available information, it may provide evidence-grounded guidance and suggest appropriate next steps for discussion with a healthcare professional. Presenting this information in understandable language may help patients follow their medical history and prepare more effectively for consultations.

F. Discussion

The main contribution of the proposed framework is the integration of multimodal medical interpretation, multi-disease analysis, longitudinal health monitoring, and explainable information retrieval within a unified patient-centered platform. This integrated approach is intended to provide broader context for understanding medical information and support informed healthcare decisions [1], [2].

Therefore, the framework's actual effectiveness must be established through experiments using reliable reference labels, appropriate evaluation metrics, and, where possible, expert-verified assessments. Measures such as precision, recall, F1-score, sensitivity, specificity, medical information extraction accuracy, and the factual grounding of generated explanations should be evaluated before drawing conclusions about its performance or clinical usefulness [1], [2], [6]. Expected deliverables are the unified dataset, the extraction and integration software, the multimodal interpretation models, the longitudinal analysis module, the RAG explanation layer and a prototype patient dashboard.

XIV. Limitations

However, several challenges must be addressed. Medical reports may contain inconsistent terminology, missing values, different measurement units, or incomplete historical information. Incorrect patient-record matching may also affect longitudinal analysis. In addition, pretrained models can produce inaccurate or unsupported interpretations, and performance may vary across diseases and data sources. The proposed real-world dataset collaboration with Lothe Lab, targeting approximately 2,261 patients, remains subject to data availability, authorization, and applicable privacy requirements. Retrieved references must be curated and kept current, and the system must be used only as decision support with appropriate clinical oversight [6].

XV. Conclusion and Future Work

This paper presents an explainable multimodal AI framework that integrates laboratory reports, clinical notes, symptoms, prescriptions and supported medical images, analyzes multiple diseases, tracks changes in patient health over time, and grounds its explanations in trusted medical references through Retrieval-Augmented Generation. The framework is intended to support patients and healthcare professionals rather than replace clinical judgment [6], and its effectiveness will be established through the statistical analysis protocol and target values described above once the dataset and models are complete. Six selected optimization techniques are specified to improve extraction, retrieval, trend analysis and the reliability of model confidence.

Future work includes extending the framework to more diseases and imaging modalities such as MRI and DICOM studies, multilingual patient-friendly explanations, real-time health monitoring, clinician-in-the-loop validation, and prospective evaluation with appropriate ethical approval.

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